SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	**CSA1223**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	CS & BIOSCIENCE	**Date:**	

83%

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization
   A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

2. Cloud Server Memory Bottleneck
   A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

3. Autonomous Vehicle Data Processing
   An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

4. AI Inference Edge Device
   An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

5. High-Performance Gaming Console
   A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct

I D. Sujan Bai (M25[illegible] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

D. Sujan Bai
Signature of the student:

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)	
Q1	1	1.25	1.25	0.75	0.5	4.75
Q2	1	1	1.25	0.75	0.5	4.5
Q3	0.5	1.25	1	0.75	0.5	4
Q4	1	1	1	0.5	0.5	4
Q5	0.5	1	1	0.5	0.5	3.5
						20.75

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints(1)	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem(1.25)	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards(1.25)	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data(0.75)	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation(0.5)	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Q1. Smartphone Performance Optimization
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
Answer

Industry Problem

Modern smartphones run multiple applications simultaneously. During app switching, the processor must quickly access data and instructions. If memory access is slow, users experience lag and delayed response.

Role of Memory Components

L1 Cache

- Closest to CPU.
- Smallest and fastest cache.
- Stores frequently used instructions and data.
- Access time: Very low latency.

L2 Cache

- Larger than L1.
- Stores recently used data.
- Acts as backup when data is unavailable in L1.

L3 Cache

- Shared among CPU cores.
- Reduces repeated DRAM accesses.
- Improves multitasking performance.

DRAM

- Main memory of smartphone.
- Large storage capacity.
- Slower than cache memory.
- Holds active applications and operating system data.

Proposed Hierarchical Memory Redesign

1. Increase L1 cache size moderately.
2. Expand shared L3 cache.
3. Use LPDDR5X DRAM for higher bandwidth.
4. Implement intelligent cache prefetching.
5. Prioritize frequently used applications in cache.

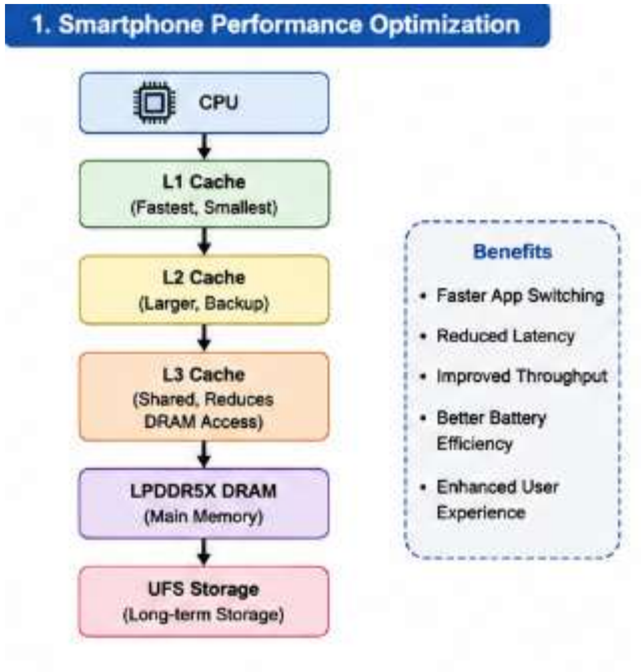
Benefits

- Faster app switching.
- Reduced latency.
- Improved throughput.
- Better battery efficiency.
- Enhanced user experience.

Analysis

The redesigned hierarchy reduces DRAM access frequency by keeping frequently accessed data in cache levels. This significantly improves application responsiveness.

Diagram



Q2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

Answer

Industry Problem

Cloud servers handle millions of database requests. Slow memory access creates query delays and affects service quality.

Cache Hierarchy

Advantages

- Very fast access.
- Reduces CPU waiting time.
- Improves database performance.

Disadvantages

- Limited capacity.
- Higher cost per bit.

Virtual Memory Paging

Advantages

- Supports large applications.
- Efficient memory utilization.

Disadvantages

- Page faults increase latency.
- Disk access is much slower than cache.

Comparison

Feature	Cache Hierarchy	Virtual Memory
Speed	Very High	Lower
Capacity	Small	Large
Latency	Low	High
Purpose	Fast access	Memory expansion

Recommended Improvements

1. Increase L3 cache capacity.
2. Use high-speed DDR5 DRAM.
3. Reduce paging activity.
4. Employ NVMe SSD for swap storage.
5. Optimize database caching algorithms.

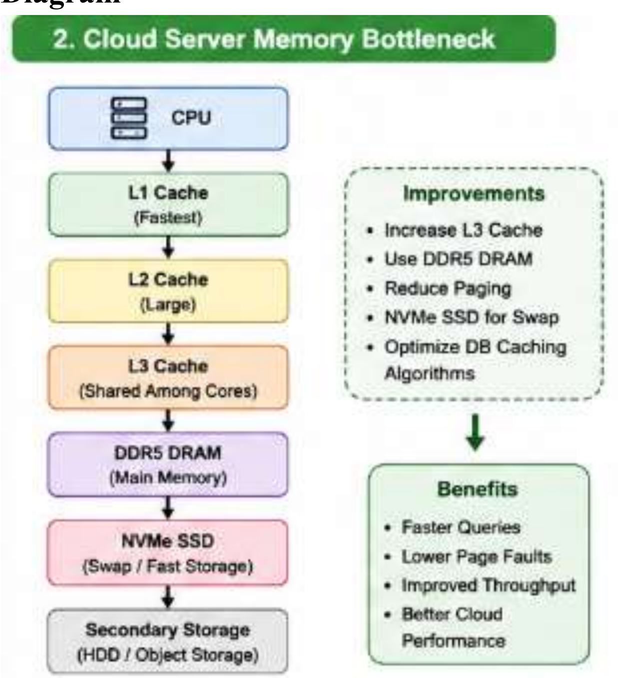
Benefits

- Faster query execution.
- Lower page faults.
- Improved server throughput.
- Better cloud performance.

Analysis

Database data should remain in cache and DRAM whenever possible. Reducing paging significantly decreases query response time.

Diagram



Q3. Autonomous Vehicle Data Processing

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Answer

Industry Problem

Autonomous vehicles continuously process data from cameras, LiDAR, radar, and sensors. Real-time decision-making requires extremely fast memory access.

Memory Technologies

SRAM

- Fastest memory.
- Used in cache.
- Low latency.
- Expensive.

DRAM

- Large capacity.
- Moderate speed.
- Stores active datasets.

MRAM

- Non-volatile memory.
- Fast read/write operations.
- Retains data without power.

Proposed Memory Hierarchy

1. SRAM for L1/L2 caches.
2. DRAM for sensor buffering.
3. MRAM for critical system data.
4. SSD for long-term storage.

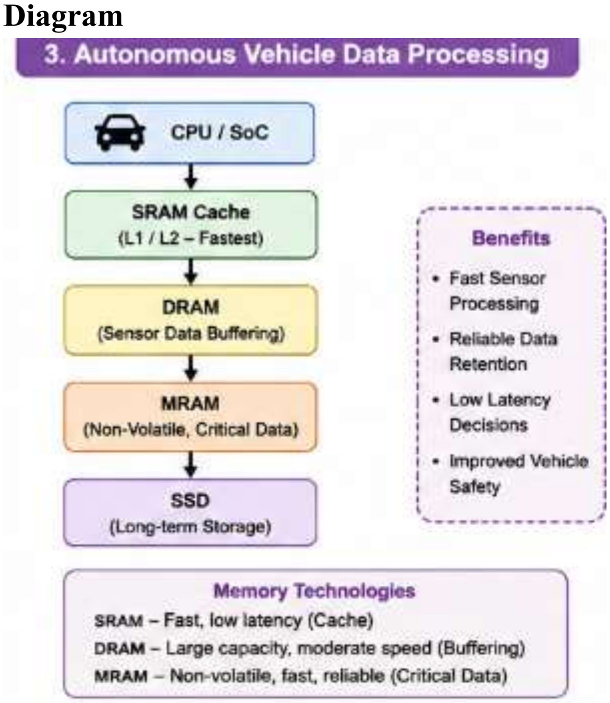
Benefits

- Fast sensor processing.
- Reliable data retention.
- Low latency decisions.
- Improved vehicle safety.

Analysis

SRAM provides instant access to critical instructions, DRAM handles large sensor streams, and MRAM preserves essential information during power interruptions.

Diagram



Q4. AI Inference Edge Device
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

Industry Problem
Edge AI devices perform local inference and must load machine learning models quickly while consuming minimal power.
NAND Flash
Advantages
- High storage capacity.
- Non-volatile.
- Low cost.

Disadvantages
- Slow access speed.

DRAM
Advantages
- Fast access.
- Suitable for active AI models.

Disadvantages
- Volatile memory.
- Higher power usage.

RRAM
Advantages
- Non-volatile.
- Low power consumption.
- Fast switching speed.

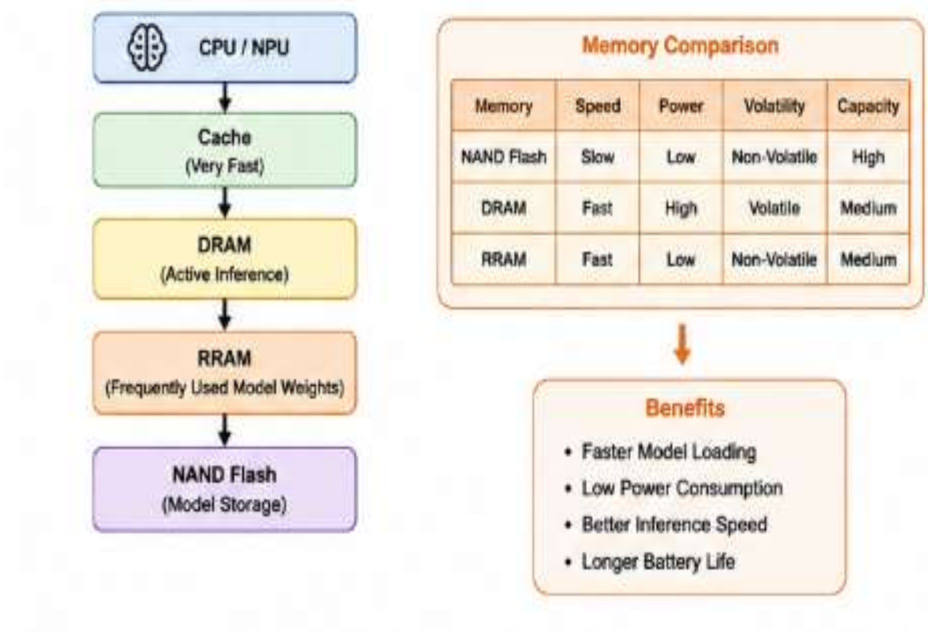
Disadvantages
- Limited commercial adoption.

Proposed Hierarchy
1. NAND Flash stores AI models.
2. RRAM stores frequently used model weights.
3. DRAM handles active inference tasks.
4. Cache stores immediate computations.

Benefits
- Faster model loading.
- Reduced energy consumption.
- Better inference speed.
- Longer battery life.

Analysis
Combining NAND Flash, RRAM, and DRAM creates a balanced architecture that improves performance while minimizing power consumption.

Diagram



Q5. High-Performance Gaming Console

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Industry Problem

Modern games load high-resolution textures and complex environments. Slow memory access causes frame drops and gameplay interruptions.

L3 Cache

Advantages

- Very low latency.
- Shared among processor cores.
- Improves game responsiveness.

Disadvantages

- Limited capacity.

DRAM

Advantages

- Large storage capacity.
- Stores game assets and textures.

Disadvantages

- Higher latency than cache.

Proposed Enhancements

1. Increase L3 cache size.
2. Use GDDR6/DDR5 memory.
3. Implement texture prefetching.
4. Improve memory bandwidth.
5. Optimize cache management algorithms.

Benefits

- Smooth gameplay.
- Faster scene loading.
- Reduced frame drops.
- Higher frame rates.

Analysis

A larger L3 cache reduces expensive DRAM accesses, while high-bandwidth DRAM ensures rapid asset loading during gameplay.

Diagram